```
Mounted at /content/drive
Seeds: [42, 52, 62, 72, 82]

Checking dataset paths...
Dataset found at:
/content/drive/MyDrive/Bone_Fracture_Binary_Classification

================ DATASET INFORMATION ================
Training images   : 9246
Validation images : 829
Testing images    : 506
Classes           : ['fractured', 'not fractured']
Class to index    : {'fractured': 0, 'not fractured': 1}

Using Device: cuda
GPU: Tesla T4

================ MODEL ARCHITECTURE ================
Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_
100%|██████████| 20.5M/20.5M [00:00<00:00, 90.4MB/s]
EfficientNet_8(
  (backbone): EfficientNet(
    (features): Sequential(
      (0): Conv2dNormActivation(
        (0): Conv2d(3, 32, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=False)
        (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        (2): SiLU(inplace=True)
      )
      (1): Sequential(
        (0): MBConv(
          (block): Sequential(
            (0): Conv2dNormActivation(
              (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
              (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (1): SqueezeExcitation(
              (avgpool): AdaptiveAvgPool2d(output_size=1)
              (fc1): Conv2d(32, 8, kernel_size=(1, 1), stride=(1, 1))
              (fc2): Conv2d(8, 32, kernel_size=(1, 1), stride=(1, 1))
              (activation): SiLU(inplace=True)
              (scale_activation): Sigmoid()
            )
            (2): Conv2dNormActivation(
              (0): Conv2d(32, 16, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            )
          )
          (stochastic_depth): StochasticDepth(p=0.0, mode=row)
        )
      )
      (2): Sequential(
        (0): MBConv(
          (block): Sequential(
            (0): Conv2dNormActivation(
              (0): Conv2d(16, 96, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (1): Conv2dNormActivation(
              (0): Conv2d(96, 96, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups=96, bias=False)
              (1): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (2): SqueezeExcitation(
              (avgpool): AdaptiveAvgPool2d(output_size=1)
              (fc1): Conv2d(96, 4, kernel_size=(1, 1), stride=(1, 1))
              (fc2): Conv2d(4, 96, kernel_size=(1, 1), stride=(1, 1))
              (activation): SiLU(inplace=True)
              (scale_activation): Sigmoid()
            )
            (3): Conv2dNormActivation(
              (0): Conv2d(96, 24, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(24, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            )
          )
          (stochastic_depth): StochasticDepth(p=0.0125, mode=row)
        )
        (1): MBConv(
          (block): Sequential(
            (0): Conv2dNormActivation(
              (0): Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
              (2): SiLU(inplace=True)
            )
            (1): Conv2dNormActivation(
```

```
          (0): Conv2d(144, 144, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=144, bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(144, 6, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(6, 144, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(144, 24, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(24, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.025, mode=row)
    )
  )
  (3): Sequential(
    (0): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(144, 144, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2), groups=144, bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(144, 6, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(6, 144, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(144, 40, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(40, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.037500000000000006, mode=row)
    )
    (1): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(40, 240, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(240, 240, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=240, bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(240, 10, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(10, 240, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(240, 40, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(40, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.05, mode=row)
    )
  )
  (4): Sequential(
    (0): MBConv(
      (block): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(40, 240, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(240, 240, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups=240, bias=False)
          (1): BatchNorm2d(240, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
```

```
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(240, 10, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(10, 240, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(240, 80, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(80, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.0625, mode=row)
      )
      (1): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(80, 480, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(480, 480, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=480, bias=False)
            (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(480, 20, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(20, 480, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(480, 80, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(80, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.07500000000000001, mode=row)
      )
      (2): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(80, 480, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(480, 480, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=480, bias=False)
            (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(480, 20, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(20, 480, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(480, 80, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(80, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.08750000000000001, mode=row)
      )
    )
    (5): Sequential(
      (0): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(80, 480, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(480, 480, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=480, bias=False)
            (1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(480, 20, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(20, 480, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
```

```
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(480, 112, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(112, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.1, mode=row)
      )
      (1): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(112, 672, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(672, 672, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=672, bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(672, 28, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(28, 672, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(672, 112, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(112, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.1125, mode=row)
      )
      (2): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(112, 672, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(672, 672, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=672, bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(672, 28, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(28, 672, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(672, 112, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(112, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.125, mode=row)
      )
    )
    (6): Sequential(
      (0): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(112, 672, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(672, 672, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2), groups=672, bias=False)
            (1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(672, 28, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(28, 672, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(672, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
```

```
        (stochastic_depth): StochasticDepth(p=0.1375, mode=row)
      )
      (1): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(1152, 1152, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=1152, bias=False)
            (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(1152, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.15000000000000002, mode=row)
      )
      (2): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(1152, 1152, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=1152, bias=False)
            (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(1152, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.1625, mode=row)
      )
      (3): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (1): Conv2dNormActivation(
            (0): Conv2d(1152, 1152, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=1152, bias=False)
            (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
            (2): SiLU(inplace=True)
          )
          (2): SqueezeExcitation(
            (avgpool): AdaptiveAvgPool2d(output_size=1)
            (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
            (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
            (activation): SiLU(inplace=True)
            (scale_activation): Sigmoid()
          )
          (3): Conv2dNormActivation(
            (0): Conv2d(1152, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          )
        )
        (stochastic_depth): StochasticDepth(p=0.17500000000000002, mode=row)
      )
    )
    (7): Sequential(
      (0): MBConv(
        (block): Sequential(
          (0): Conv2dNormActivation(
            (0): Conv2d(192, 1152, kernel_size=(1, 1), stride=(1, 1), bias=False)
```

```
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(1152, 1152, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=1152, bias=False)
          (1): BatchNorm2d(1152, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
          (2): SiLU(inplace=True)
        )
        (2): SqueezeExcitation(
          (avgpool): AdaptiveAvgPool2d(output_size=1)
          (fc1): Conv2d(1152, 48, kernel_size=(1, 1), stride=(1, 1))
          (fc2): Conv2d(48, 1152, kernel_size=(1, 1), stride=(1, 1))
          (activation): SiLU(inplace=True)
          (scale_activation): Sigmoid()
        )
        (3): Conv2dNormActivation(
          (0): Conv2d(1152, 320, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(320, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
        )
      )
      (stochastic_depth): StochasticDepth(p=0.1875, mode=row)
    )
  )
  (8): Conv2dNormActivation(
    (0): Conv2d(320, 1280, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (1): BatchNorm2d(1280, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): SiLU(inplace=True)
  )
)
(avgpool): AdaptiveAvgPool2d(output_size=1)
(classifier): Identity()
)
(fc1): Linear(in_features=1280, out_features=8, bias=True)
(tanh): Tanh()
(fc2): Linear(in_features=8, out_features=1, bias=True)
)


================================================================================
STARTING TRAINING FOR SEED 42
================================================================================

Random Seed Set To: 42
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 1

Seed 42 | Epoch 1/10 | Train Loss: 0.4114 | Val Loss: 0.2924 | Train Acc: 88.62% | Val Acc: 95.54%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 2

Seed 42 | Epoch 2/10 | Train Loss: 0.2457 | Val Loss: 0.2473 | Train Acc: 98.97% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 3

Seed 42 | Epoch 3/10 | Train Loss: 0.2148 | Val Loss: 0.2140 | Train Acc: 99.32% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 4

Seed 42 | Epoch 4/10 | Train Loss: 0.1985 | Val Loss: 0.2104 | Train Acc: 99.70% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 5

Seed 42 | Epoch 5/10 | Train Loss: 0.1965 | Val Loss: 0.2060 | Train Acc: 99.62% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 6

Seed 42 | Epoch 6/10 | Train Loss: 0.1935 | Val Loss: 0.2050 | Train Acc: 99.76% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 7/10 | Train Loss: 0.1920 | Val Loss: 0.2099 | Train Acc: 99.73% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 42, Epoch 8

Seed 42 | Epoch 8/10 | Train Loss: 0.1915 | Val Loss: 0.2033 | Train Acc: 99.74% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 42 | Epoch 9/10 | Train Loss: 0.1913 | Val Loss: 0.2041 | Train Acc: 99.77% | Val Acc: 98.31%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
```

```
Seed 42 | Epoch 10/10 | Train Loss: 0.1915 | Val Loss: 0.2037 | Train Acc: 99.71% | Val Acc: 98.31%


================================================================================
SEED 42 TRAINING COMPLETE
Time taken: 41.27 minutes
Best Validation Loss: 0.203348
Model saved: /content/best_model_seed_42.pth
================================================================================

Evaluating Seed 42 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 42 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.891845   98.311218 98.023715
Sensitivity (%) 99.913793   99.186992 98.507463
Specificity (%) 99.869735   97.032641 97.478992
  Precision (%) 99.870745   97.991968 97.777778
   F1-Score (%) 99.892265   98.585859 98.141264
 Error Rate (%)  0.108155    1.688782  1.976285
        ROC-AUC  0.999960    0.997732  0.997539


================================================================================
STARTING TRAINING FOR SEED 52
================================================================================

Random Seed Set To: 52
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 1

Seed 52 | Epoch 1/10 | Train Loss: 0.3440 | Val Loss: 0.2406 | Train Acc: 89.48% | Val Acc: 93.61%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 2

Seed 52 | Epoch 2/10 | Train Loss: 0.1530 | Val Loss: 0.1827 | Train Acc: 98.98% | Val Acc: 96.86%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 3

Seed 52 | Epoch 3/10 | Train Loss: 0.1316 | Val Loss: 0.1408 | Train Acc: 99.24% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 4

Seed 52 | Epoch 4/10 | Train Loss: 0.1176 | Val Loss: 0.1387 | Train Acc: 99.60% | Val Acc: 98.79%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 5

Seed 52 | Epoch 5/10 | Train Loss: 0.1154 | Val Loss: 0.1371 | Train Acc: 99.70% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 6

Seed 52 | Epoch 6/10 | Train Loss: 0.1144 | Val Loss: 0.1340 | Train Acc: 99.65% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 7

Seed 52 | Epoch 7/10 | Train Loss: 0.1129 | Val Loss: 0.1339 | Train Acc: 99.70% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 8

Seed 52 | Epoch 8/10 | Train Loss: 0.1126 | Val Loss: 0.1338 | Train Acc: 99.72% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 52, Epoch 9

Seed 52 | Epoch 9/10 | Train Loss: 0.1109 | Val Loss: 0.1334 | Train Acc: 99.77% | Val Acc: 98.91%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 52 | Epoch 10/10 | Train Loss: 0.1115 | Val Loss: 0.1336 | Train Acc: 99.74% | Val Acc: 98.91%


================================================================================
SEED 52 TRAINING COMPLETE
Time taken: 8.58 minutes
Best Validation Loss: 0.133441
```

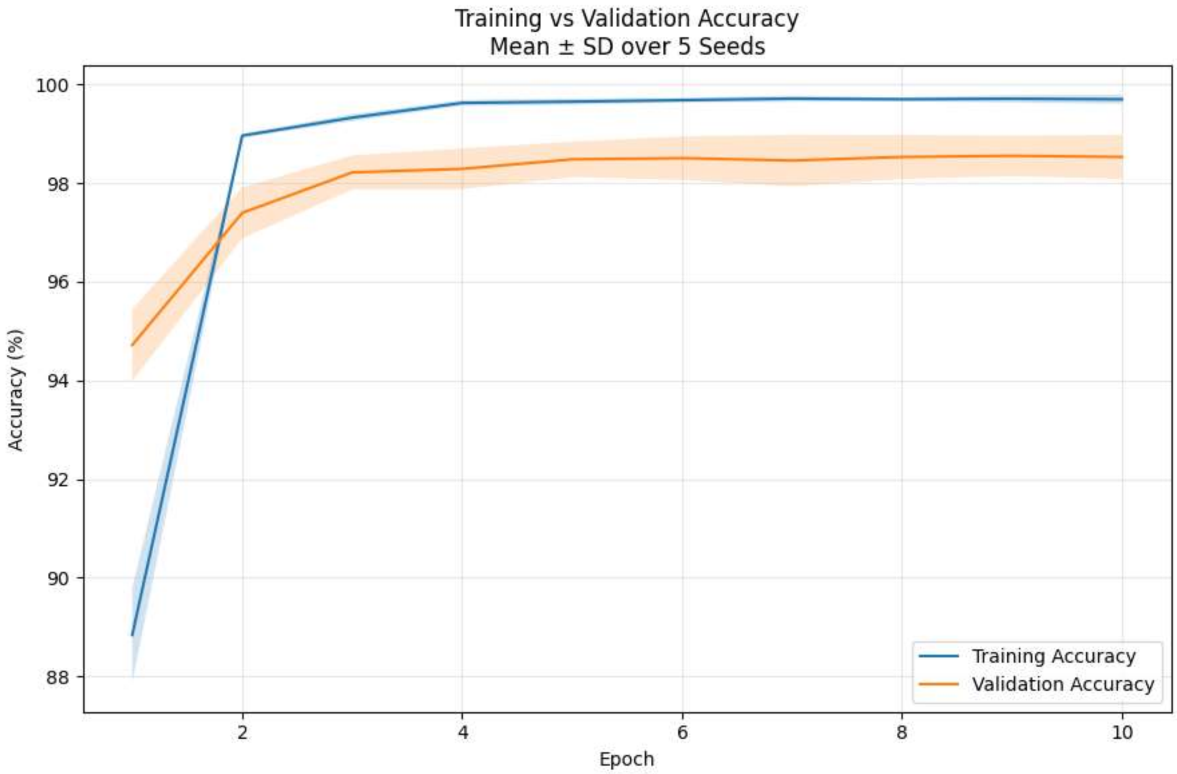
```
Model saved: /content/best_model_seed_52.pth
================================================================================

Evaluating Seed 52 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 52 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.891845   98.914355 98.616601
Sensitivity (%) 99.892241   99.390244 98.880597
Specificity (%) 99.891446   98.219585 98.319328
  Precision (%) 99.892241   98.787879 98.513011
   F1-Score (%) 99.892241   99.088146 98.696462
 Error Rate (%)  0.108155    1.085645  1.383399
        ROC-AUC  0.999984    0.998752  0.997350


================================================================================
STARTING TRAINING FOR SEED 62
================================================================================

Random Seed Set To: 62
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 1

Seed 62 | Epoch 1/10 | Train Loss: 0.4129 | Val Loss: 0.3078 | Train Acc: 90.14% | Val Acc: 95.17%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 2

Seed 62 | Epoch 2/10 | Train Loss: 0.2480 | Val Loss: 0.2629 | Train Acc: 98.99% | Val Acc: 96.86%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 3

Seed 62 | Epoch 3/10 | Train Loss: 0.2175 | Val Loss: 0.2324 | Train Acc: 99.30% | Val Acc: 97.83%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 4

Seed 62 | Epoch 4/10 | Train Loss: 0.2010 | Val Loss: 0.2276 | Train Acc: 99.58% | Val Acc: 97.71%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 5

Seed 62 | Epoch 5/10 | Train Loss: 0.1992 | Val Loss: 0.2212 | Train Acc: 99.60% | Val Acc: 97.95%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 6

Seed 62 | Epoch 6/10 | Train Loss: 0.1957 | Val Loss: 0.2209 | Train Acc: 99.68% | Val Acc: 98.07%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 7

Seed 62 | Epoch 7/10 | Train Loss: 0.1945 | Val Loss: 0.2208 | Train Acc: 99.69% | Val Acc: 98.07%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 8

Seed 62 | Epoch 8/10 | Train Loss: 0.1949 | Val Loss: 0.2192 | Train Acc: 99.69% | Val Acc: 97.95%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 62, Epoch 9

Seed 62 | Epoch 9/10 | Train Loss: 0.1948 | Val Loss: 0.2181 | Train Acc: 99.65% | Val Acc: 98.07%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 62 | Epoch 10/10 | Train Loss: 0.1952 | Val Loss: 0.2184 | Train Acc: 99.59% | Val Acc: 97.95%


================================================================================
SEED 62 TRAINING COMPLETE
Time taken: 8.72 minutes
Best Validation Loss: 0.218066
Model saved: /content/best_model_seed_62.pth
================================================================================

Evaluating Seed 62 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
```

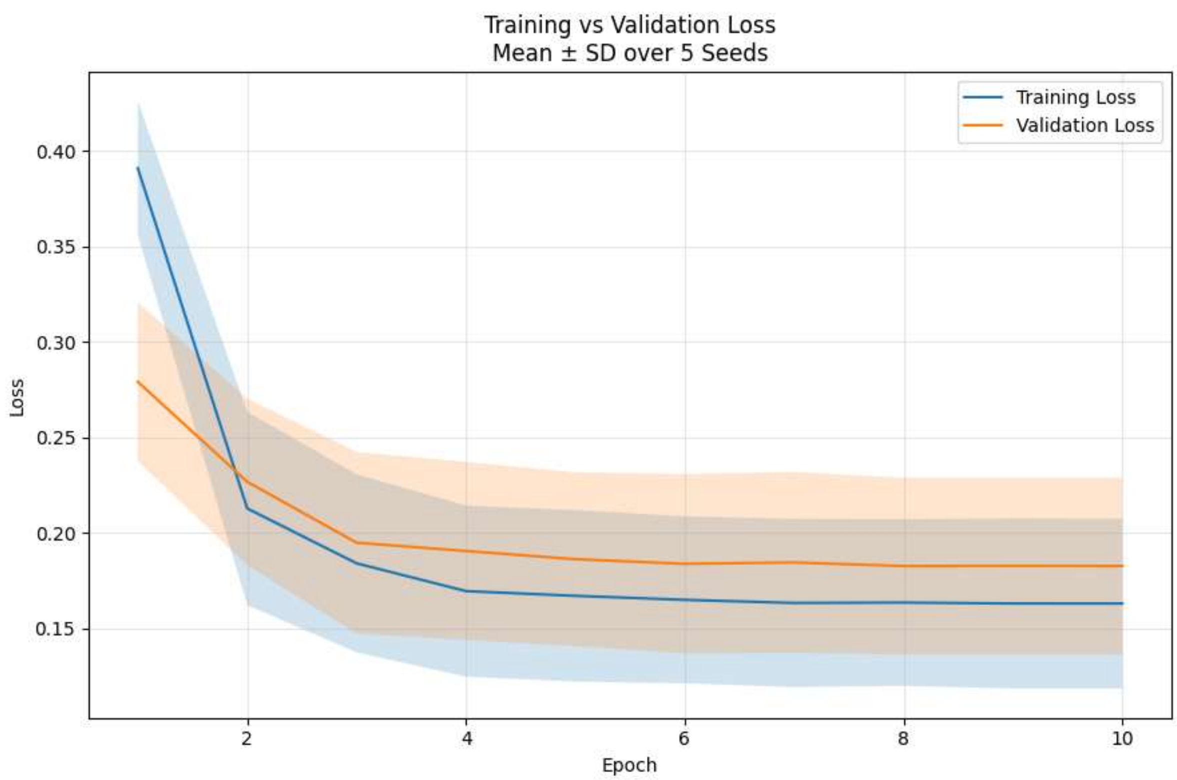
```
==========================================================================================
SEED 62 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.762059   98.069964 97.430830
Sensitivity (%) 99.719828   98.170732 96.641791
Specificity (%) 99.804603   97.922849 98.319328
  Precision (%) 99.805867   98.571429 98.479087
   F1-Score (%) 99.762829   98.370672 97.551789
 Error Rate (%)  0.237941    1.930036  2.569170
        ROC-AUC  0.999980    0.998197  0.997100


================================================================================
STARTING TRAINING FOR SEED 72
================================================================================

Random Seed Set To: 72
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 1

Seed 72 | Epoch 1/10 | Train Loss: 0.4229 | Val Loss: 0.3240 | Train Acc: 88.18% | Val Acc: 94.57%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 2

Seed 72 | Epoch 2/10 | Train Loss: 0.2549 | Val Loss: 0.2642 | Train Acc: 98.96% | Val Acc: 97.95%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 3

Seed 72 | Epoch 3/10 | Train Loss: 0.2216 | Val Loss: 0.2399 | Train Acc: 99.33% | Val Acc: 97.95%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 4

Seed 72 | Epoch 4/10 | Train Loss: 0.2066 | Val Loss: 0.2336 | Train Acc: 99.57% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 5

Seed 72 | Epoch 5/10 | Train Loss: 0.2034 | Val Loss: 0.2292 | Train Acc: 99.60% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 6

Seed 72 | Epoch 6/10 | Train Loss: 0.2009 | Val Loss: 0.2267 | Train Acc: 99.64% | Val Acc: 98.19%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 7

Seed 72 | Epoch 7/10 | Train Loss: 0.1994 | Val Loss: 0.2257 | Train Acc: 99.63% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 72, Epoch 8

Seed 72 | Epoch 8/10 | Train Loss: 0.1992 | Val Loss: 0.2249 | Train Acc: 99.64% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 72 | Epoch 9/10 | Train Loss: 0.1998 | Val Loss: 0.2258 | Train Acc: 99.60% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 72 | Epoch 10/10 | Train Loss: 0.1993 | Val Loss: 0.2256 | Train Acc: 99.63% | Val Acc: 98.43%


================================================================================
SEED 72 TRAINING COMPLETE
Time taken: 8.62 minutes
Best Validation Loss: 0.224943
Model saved: /content/best_model_seed_72.pth
================================================================================

Evaluating Seed 72 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 72 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.826952   98.431846 98.023715
Sensitivity (%) 99.784483   98.780488 97.761194
Specificity (%) 99.869735   97.922849 98.319328
  Precision (%) 99.870578   98.580122 98.496241
   F1-Score (%) 99.827512   98.680203 98.127341
```

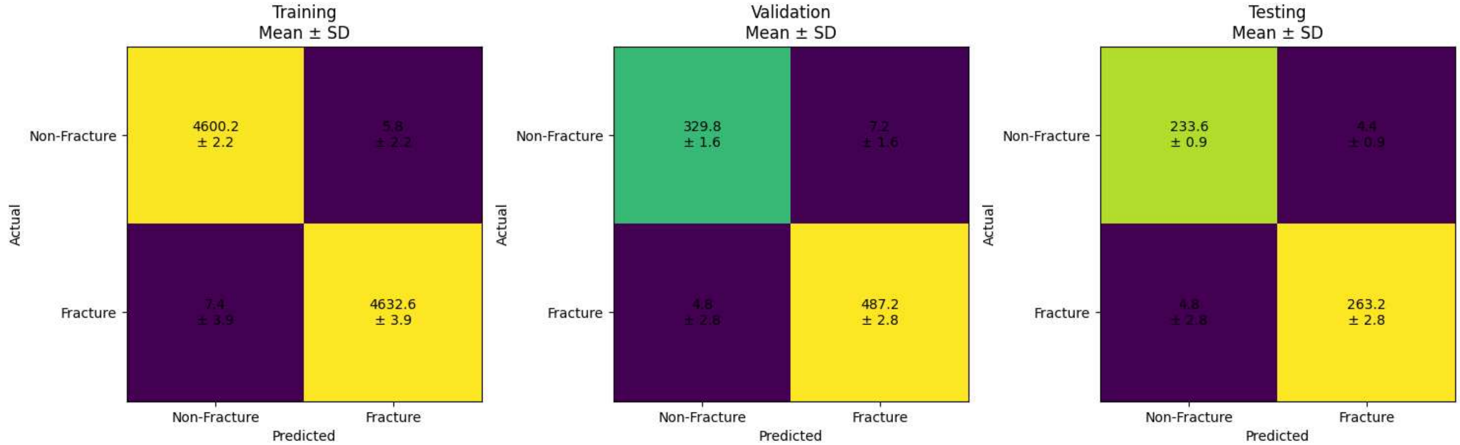
```
  Error Rate (%)  0.173048    1.568154  1.976285
         ROC-AUC  0.999836    0.997768  0.997476


================================================================================
STARTING TRAINING FOR SEED 82
================================================================================

Random Seed Set To: 82
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 1

Seed 82 | Epoch 1/10 | Train Loss: 0.3627 | Val Loss: 0.2302 | Train Acc: 87.78% | Val Acc: 94.69%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 2

Seed 82 | Epoch 2/10 | Train Loss: 0.1621 | Val Loss: 0.1764 | Train Acc: 98.89% | Val Acc: 97.47%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 3

Seed 82 | Epoch 3/10 | Train Loss: 0.1344 | Val Loss: 0.1468 | Train Acc: 99.44% | Val Acc: 98.43%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 4

Seed 82 | Epoch 4/10 | Train Loss: 0.1235 | Val Loss: 0.1419 | Train Acc: 99.68% | Val Acc: 98.55%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 5

Seed 82 | Epoch 5/10 | Train Loss: 0.1204 | Val Loss: 0.1372 | Train Acc: 99.73% | Val Acc: 98.67%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 6

Seed 82 | Epoch 6/10 | Train Loss: 0.1199 | Val Loss: 0.1321 | Train Acc: 99.66% | Val Acc: 99.03%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 7

Seed 82 | Epoch 7/10 | Train Loss: 0.1174 | Val Loss: 0.1320 | Train Acc: 99.81% | Val Acc: 99.03%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Checkpoint saved: Seed 82, Epoch 8

Seed 82 | Epoch 8/10 | Train Loss: 0.1190 | Val Loss: 0.1315 | Train Acc: 99.70% | Val Acc: 99.03%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 82 | Epoch 9/10 | Train Loss: 0.1179 | Val Loss: 0.1318 | Train Acc: 99.74% | Val Acc: 99.03%
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(
Seed 82 | Epoch 10/10 | Train Loss: 0.1173 | Val Loss: 0.1316 | Train Acc: 99.82% | Val Acc: 99.03%


================================================================================
SEED 82 TRAINING COMPLETE
Time taken: 8.60 minutes
Best Validation Loss: 0.131487
Model saved: /content/best_model_seed_82.pth
================================================================================

Evaluating Seed 82 on Training / Validation / Testing...
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
  warnings.warn(

==========================================================================================
SEED 82 PERFORMANCE
==========================================================================================
         Metric  Training  Validation   Testing
   Accuracy (%) 99.913476   99.034982 98.814229
Sensitivity (%) 99.892241   99.593496 99.253731
Specificity (%) 99.934868   98.219585 98.319328
  Precision (%) 99.935317   98.790323 98.518519
   F1-Score (%) 99.913775   99.190283 98.884758
 Error Rate (%)  0.086524    0.965018  1.185771
        ROC-AUC  0.999992    0.999234  0.998667


==========================================================================================
ALL FIVE SEEDS COMPLETED
==========================================================================================
Seeds completed: [42, 52, 62, 72, 82]
Total time: 83.00 minutes
```

```
==========================================================================================
PER-SEED PERFORMANCE RESULTS
==========================================================================================
 Seed    Dataset          Metric     Value
   42   Training    Accuracy (%) 99.891845
   42   Training Sensitivity (%) 99.913793
   42   Training Specificity (%) 99.869735
   42   Training   Precision (%) 99.870745
   42   Training    F1-Score (%) 99.892265
   42   Training  Error Rate (%)  0.108155
   42   Training         ROC-AUC  0.999960
   42 Validation    Accuracy (%) 98.311218
   42 Validation Sensitivity (%) 99.186992
   42 Validation Specificity (%) 97.032641
   42 Validation   Precision (%) 97.991968
   42 Validation    F1-Score (%) 98.585859
   42 Validation  Error Rate (%)  1.688782
   42 Validation         ROC-AUC  0.997732
   42    Testing    Accuracy (%) 98.023715
   42    Testing Sensitivity (%) 98.507463
   42    Testing Specificity (%) 97.478992
   42    Testing   Precision (%) 97.777778
   42    Testing    F1-Score (%) 98.141264
   42    Testing  Error Rate (%)  1.976285
   42    Testing         ROC-AUC  0.997539
   52   Training    Accuracy (%) 99.891845
   52   Training Sensitivity (%) 99.892241
   52   Training Specificity (%) 99.891446
   52   Training   Precision (%) 99.892241
   52   Training    F1-Score (%) 99.892241
   52   Training  Error Rate (%)  0.108155
   52   Training         ROC-AUC  0.999984
   52 Validation    Accuracy (%) 98.914355
   52 Validation Sensitivity (%) 99.390244
   52 Validation Specificity (%) 98.219585
   52 Validation   Precision (%) 98.787879
   52 Validation    F1-Score (%) 99.088146
   52 Validation  Error Rate (%)  1.085645
   52 Validation         ROC-AUC  0.998752
   52    Testing    Accuracy (%) 98.616601
   52    Testing Sensitivity (%) 98.880597
   52    Testing Specificity (%) 98.319328
   52    Testing   Precision (%) 98.513011
   52    Testing    F1-Score (%) 98.696462
   52    Testing  Error Rate (%)  1.383399
   52    Testing         ROC-AUC  0.997350
   62   Training    Accuracy (%) 99.762059
   62   Training Sensitivity (%) 99.719828
   62   Training Specificity (%) 99.804603
   62   Training   Precision (%) 99.805867
   62   Training    F1-Score (%) 99.762829
   62   Training  Error Rate (%)  0.237941
   62   Training         ROC-AUC  0.999980
   62 Validation    Accuracy (%) 98.069964
   62 Validation Sensitivity (%) 98.170732
   62 Validation Specificity (%) 97.922849
   62 Validation   Precision (%) 98.571429
   62 Validation    F1-Score (%) 98.370672
   62 Validation  Error Rate (%)  1.930036
   62 Validation         ROC-AUC  0.998197
   62    Testing    Accuracy (%) 97.430830
   62    Testing Sensitivity (%) 96.641791
   62    Testing Specificity (%) 98.319328
   62    Testing   Precision (%) 98.479087
   62    Testing    F1-Score (%) 97.551789
   62    Testing  Error Rate (%)  2.569170
   62    Testing         ROC-AUC  0.997100
   72   Training    Accuracy (%) 99.826952
   72   Training Sensitivity (%) 99.784483
   72   Training Specificity (%) 99.869735
   72   Training   Precision (%) 99.870578
   72   Training    F1-Score (%) 99.827512
   72   Training  Error Rate (%)  0.173048
   72   Training         ROC-AUC  0.999836
   72 Validation    Accuracy (%) 98.431846
   72 Validation Sensitivity (%) 98.780488
   72 Validation Specificity (%) 97.922849
   72 Validation   Precision (%) 98.580122
   72 Validation    F1-Score (%) 98.680203
   72 Validation  Error Rate (%)  1.568154
   72 Validation         ROC-AUC  0.997768
   72    Testing    Accuracy (%) 98.023715
   72    Testing Sensitivity (%) 97.761194
   72    Testing Specificity (%) 98.319328
```

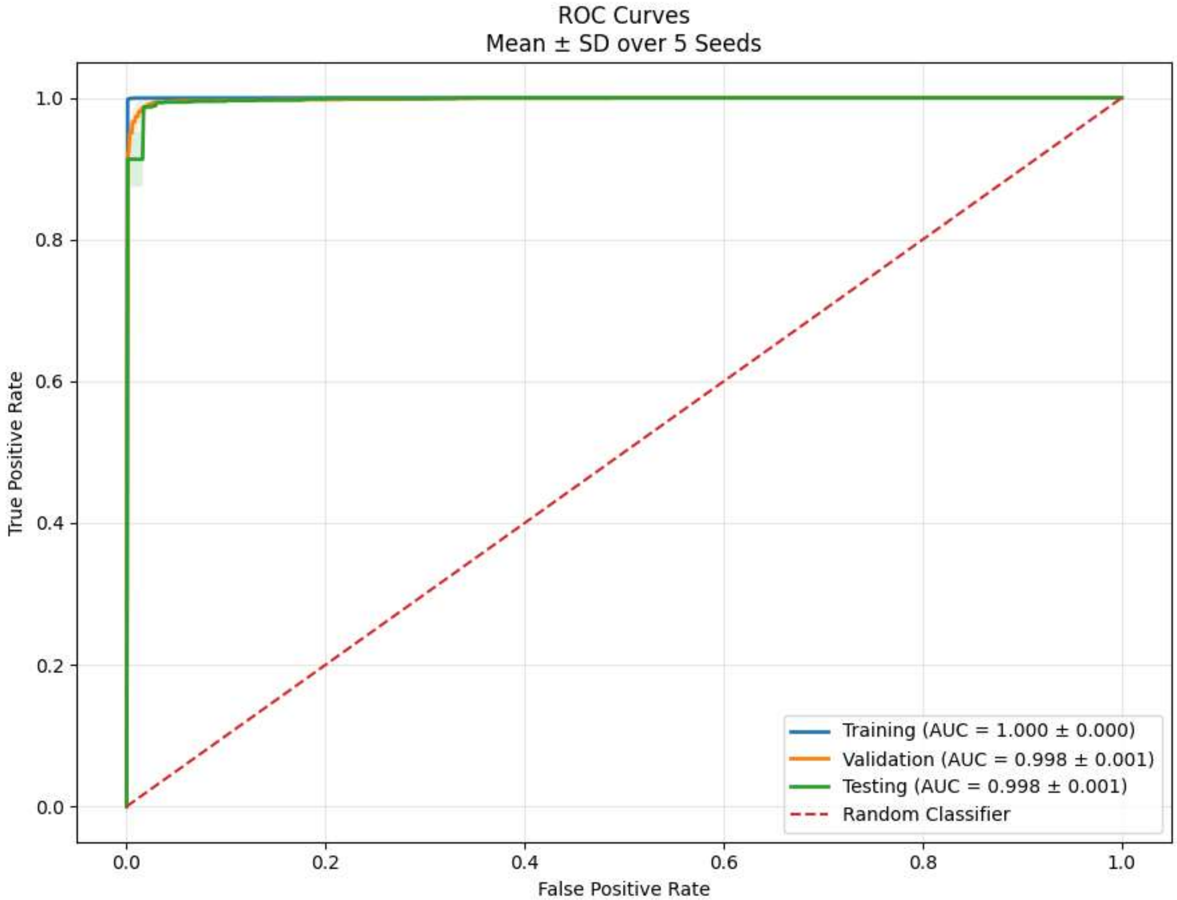
```
    72     Testing   Precision (%) 98.496241
    72     Testing    F1-Score (%) 98.127341
    72     Testing  Error Rate (%)  1.976285
    72     Testing         ROC-AUC  0.997476
    82    Training    Accuracy (%) 99.913476
    82    Training Sensitivity (%) 99.892241
    82    Training Specificity (%) 99.934868
    82    Training   Precision (%) 99.935317
    82    Training    F1-Score (%) 99.913775
    82    Training  Error Rate (%)  0.086524
    82    Training         ROC-AUC  0.999992
    82  Validation    Accuracy (%) 99.034982
    82  Validation Sensitivity (%) 99.593496
    82  Validation Specificity (%) 98.219585
    82  Validation   Precision (%) 98.790323
    82  Validation    F1-Score (%) 99.190283
    82  Validation  Error Rate (%)  0.965018
    82  Validation         ROC-AUC  0.999234
    82     Testing    Accuracy (%) 98.814229
    82     Testing Sensitivity (%) 99.253731
    82     Testing Specificity (%) 98.319328
    82     Testing   Precision (%) 98.518519
    82     Testing    F1-Score (%) 98.884758
    82     Testing  Error Rate (%)  1.185771
    82     Testing         ROC-AUC  0.998667


==========================================================================================
FINAL PERFORMANCE COMPARISON TABLE
MEAN ± SD OVER FIVE SEEDS
==========================================================================================
         Metric     Training   Validation      Testing
   Accuracy (%) 99.86 ± 0.06 98.55 ± 0.41 98.18 ± 0.55
Sensitivity (%) 99.84 ± 0.08 99.02 ± 0.56 98.21 ± 1.04
Specificity (%) 99.87 ± 0.05 97.86 ± 0.49 98.15 ± 0.38
  Precision (%) 99.87 ± 0.05 98.54 ± 0.33 98.36 ± 0.32
   F1-Score (%) 99.86 ± 0.06 98.78 ± 0.35 98.28 ± 0.53
 Error Rate (%)  0.14 ± 0.06  1.45 ± 0.41  1.82 ± 0.55
        ROC-AUC  1.00 ± 0.00  1.00 ± 0.00  1.00 ± 0.00


==========================================================================================
TIME TAKEN FOR EACH SEED
==========================================================================================
 Seed  Time (minutes)
   42     2476.387001
   52      514.625144
   62      523.200188
   72      517.138726
   82      516.053649

Total execution time: 83.00 minutes
```


Training vs Validation Accuracy
Mean ± SD over 5 Seeds
Accuracy (%)
100
98
96
94
92
90
88
2
4
6
8
10
Epoch
Training Accuracy
Validation Accuracy


Training vs Validation Loss
Mean ± SD over 5 Seeds
Loss
0.40
0.35
0.30
0.25
0.20
0.15
2
4
6
8
10
Epoch
Training Loss
Validation Loss




```
==========================================================================================
MEAN ± SD CONFUSION MATRICES
==========================================================================================

Training Confusion Matrix:
[[4600.2    5.8]
 [   7.4 4632.6]]

Training CM SD:
[[2.17 2.17]
 [3.91 3.91]]

Validation Confusion Matrix:
[[329.8   7.2]
 [  4.8 487.2]]

Validation CM SD:
[[1.64 1.64]
 [2.77 2.77]]

Testing Confusion Matrix:
[[233.6   4.4]
 [  4.8 263.2]]

Testing CM SD:
[[0.89 0.89]
 [2.77 2.77]]
```




```
==========================================================================================
ROC-AUC MEAN ± SD
==========================================================================================
Training   : 1.0000 ± 0.0001
Validation : 0.9983 ± 0.0006
Testing    : 0.9976 ± 0.0006


====================================================================================================
PERFORMANCE COMPARISON TABLE
MEAN ± SD OVER FIVE SEEDS
====================================================================================================
         Metric     Training   Validation      Testing
   Accuracy (%) 99.86 ± 0.06 98.55 ± 0.41 98.18 ± 0.55
Sensitivity (%) 99.84 ± 0.08 99.02 ± 0.56 98.21 ± 1.04
Specificity (%) 99.87 ± 0.05 97.86 ± 0.49 98.15 ± 0.38
  Precision (%) 99.87 ± 0.05 98.54 ± 0.33 98.36 ± 0.32
   F1-Score (%) 99.86 ± 0.06 98.78 ± 0.35 98.28 ± 0.53
 Error Rate (%)  0.14 ± 0.06  1.45 ± 0.41  1.82 ± 0.55
        ROC-AUC  1.00 ± 0.00  1.00 ± 0.00  1.00 ± 0.00


==========================================================================================
RESULT FILES SAVED
==========================================================================================
/content/final_mean_sd_results.csv
/content/per_seed_results.csv
/content/seed_time_results.csv


====================================================================================================
FINAL EXPERIMENT SUMMARY
====================================================================================================
Architecture:
EfficientNet-B0 → 1280 → Linear(1280→8) → Tanh → Linear(8→1) → Sigmoid

Seeds: [42, 52, 62, 72, 82]

Epochs per seed: 10

Batch size: 64

Learning rate: 0.0001
```

```
Device: cuda

Total execution time: 83.00 minutes

All processing completed successfully.
```